

Abdullah Chabaytah

514-705-5932 | a.chabaytah@mail.utoronto.ca | [linkedin.com/in/achabaytah](https://www.linkedin.com/in/achabaytah) | github.com/maniic | maniic.github.io

EDUCATION

University of Toronto, St. George

Toronto, ON

HBSc Computer Science, Math & Statistics

Expected Grad: 05/2028

- **Relevant Coursework:** *Data Structures & Algorithms, Theoretical Computation, Software Design, Systems Programming, Multivariable Calculus, Linear Algebra, Probability, Statistics & Data Analysis*

EXPERIENCE

NeedList.ORG

09/2026 - 12/2026

Incoming Full-Stack Developer Intern, AI Specialist

Toronto, Canada

- Selected for an upcoming full-stack role focused on building scalable charity technology using **TypeScript**, **NestJS**, **React**, **TanStack**, and **Firebase**

Google Developer Groups

06/2026 – Present

Software Developer

University of Toronto, St. George

- Designed and shipped a responsive website for GDG @ UofT St. George using **React**, **TypeScript**, and **HTML/CSS**, serving **1000+** students
- Built reusable **frontend** components to showcase events, resources, and initiatives, cutting page load times by **30%** through code splitting and asset optimization

Sports Up

05/2025 – 09/2025

Software Developer Intern

Toronto, Canada

- Engineered a cross-platform mobile application using **React Native** & **Firebase**, achieving **99% uptime** across **400+ users** under sustained loads of **50 requests/sec** via real-time data synchronization & query batching
- **Reduced crash rates by 35% & increased weekly active users by 20%** by refactoring backend architecture and rebuilding **UI/UX** components to eliminate state management bottlenecks
- Implemented automated end-to-end test suites and deployment quality gates, reducing regression failures across **3** major feature releases

Outlier AI

01/2025 – Present

Data Labeling Contractor

Remote

- Annotated & validated **NLP** and **computer vision** training datasets for large language models, applying domain-specific labeling schemas to improve model precision across production **machine learning** pipelines

PROJECTS

Predictive Maintenance | *Python, PyTorch, MLflow, Scikit-learn, Pandas*

- Trained **deep learning** architectures including **Bidirectional LSTM**, **Temporal CNN**, and **Transformer** using **PyTorch Lightning** to predict jet engine failure on NASA benchmark data, achieving **11.71 RMSE**
- Built end-to-end data pipelines processing sensor channels across multiple fault-mode datasets, implementing normalization, sequence windowing, and automated data quality validation

Collaborative Code Editor | *Java, Spring Boot, WebSockets, PostgreSQL, Redis, Docker*

- Built a real-time collaborative code editor using **Java 21**, **Spring Boot**, and **WebSockets**, synchronizing edits, cursors, and user presence across shared coding sessions
- Implemented a server-side **Operational Transform** engine backed by **PostgreSQL** and **Redis**, supporting reconnect/resync flows, cross-instance pub/sub, and isolated **Docker** execution sandboxes

Sentiment Forecasting | *Python, PyTorch, Hugging Face, Scikit-learn, Pandas*

- Built a **Python** quantitative forecasting pipeline achieving a **2.34 Sharpe ratio** and **-2.5% max drawdown** over **180-day** backtests by fusing an **NLP** model outputs with topological & momentum features
- Trained and evaluated **Logistic Regression**, **XGBoost**, and **Ensemble** classifiers using time-series aware cross-validation to eliminate look-ahead bias, selecting the optimal model via **Sharpe ratio** on held-out data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, Golang, R, SQL, HTML, CSS

Frameworks & Libraries: React, Node.js, TensorFlow, PyTorch, Flask, Django, Pandas, NumPy, Scikit-learn

Systems & Tools: Docker, PostgreSQL, Redis, Git, Unix/Linux, CI/CD, VS Code, PyCharm, IntelliJ